

Aditya Hansoty

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EDUCATION

University of Michigan – Ann Arbor

Aug. 2025 – May 2029

B.S. Biomedical Engineering, Minor in Computer Science | GPA: 3.73/4.0

EXPERIENCE

Prosthetics Research Intern

May 2026 – Present

Rehabilitation Robotics Lab, University of Illinois Chicago

Chicago, IL

- Designed and iterated threaded encoder caps for a powered ankle-foot prosthesis in SolidWorks, refining geometry and tolerances across 3D-printed and CNC-machined revisions.
- Controlled a powered ankle-foot prosthesis via TwinCAT and led Vicon motion-capture setup across 10 human-subject trials; built Python pipelines synchronizing EMG, ECG, and metabolic-cost data for a journal manuscript.
- Engineered a signal-agnostic AR biofeedback platform (Python, Unity, HoloLens 2) pixel-streaming live via WebSockets to a head-mounted display for real-time gait feedback; sole developer, open-sourced.
- Led first-author manuscript submitted to ICRA 2027 using OpenCV marker tracking to characterize AR display latency; found HoloLens 2 latency (120.2 ms) statistically equivalent to a wired monitor baseline (119.2 ms); designed and conducted a 2-subject pilot assessing AR-driven gait symmetry improvement.

Exoskeleton Research Intern

Sept. 2026 – Present

Locomotor Control Systems Laboratory, University of Michigan

Ann Arbor, MI

- Collecting and processing motion capture and EMG data from clinical trials evaluating a knee exoskeleton in patients with osteoarthritis and stroke.
- Redesigning hip-belt geometry and dorsal motor-housing mounts to improve comfort and eliminate actuator-induced sag during extended wear.

Cancer Biology Research Intern

June 2025 – Aug. 2025

University of Chicago Medicine

Chicago, IL

- Investigated β -tubulin isotype knockouts in bone-metastatic breast cancer cell lines using cell culture, adhesion assays, Western blotting, and confocal microscopy; presented findings to 100+ researchers at the University of Chicago Comprehensive Cancer Center Symposium.

PROJECTS

Tibial Locking Compression Plate – ASTM F382 Validation | *SolidWorks, FEA, Simulink*

Sept. 2026

- Designed a parametric 10-hole Ti-6Al-4V locking compression plate in SolidWorks with dimensions driven by global variables.
- Validated to ASTM F382 via FEA in SolidWorks Simulation: 4-point bending (500 N) yielded FOS 1.73; torsion (10 N·m) yielded FOS 2.36; three-level mesh convergence confirmed stress converged within 3%.
- Built a Simulink lumped-parameter healing model (plate and callus springs in parallel, logistic stiffness curve from literature) showing 46% plate stress reduction over 12 weeks.

EMG-Controlled Prosthetic Gripper | *SolidWorks, Arduino, Python*

May 2025 – Present

- Designed and 3D-printed a 2-finger gripper in SolidWorks, integrating an MG996R servo, MyoWare 2.0 EMG sensor, and Arduino Uno with external power regulation to eliminate servo stutter under load.
- Benchmarked two EMG classification pipelines across electrode-replacement sessions; achieved 99.4% accuracy across three forearm gestures using LDA with real-time end-to-end control.

ACTIVITIES

Biomedical Engineering Society

Aug. 2025 – Present

Professional Development Executive Board

Ann Arbor, MI

- Led 10+ technical workshops for 100+ members on SolidWorks, wet-lab techniques, and resume development.

TECHNICAL SKILLS

Engineering: SolidWorks, SolidWorks Simulation (FEA), ASTM F382, 3D Printing, CNC Machining, Arduino, Simulink, TwinCAT, Vicon Motion Capture, EMG/ECG Signal Processing, LSL

Programming: Python, C++, MATLAB, Unity

Data & ML: OpenCV, WebSockets, NumPy, Pandas, SciPy, scikit-learn